

IL-17 研究及投资机会探讨/ IL-17-targeted Therapeutics: Research and Investment Opportunities¹

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核心要点 / Key Points

从研发角度分析，IL-17A 抑制剂在 IL-17 抑制剂中最早上市，针对银屑病等一系列自免疫疾病产生了较好的疗效，且安全性较高，表现出良好的成药性。

IL-17A/F 联合抑制剂在银屑病上也取得了优异的成绩，但仍有待推广到更多适应症上。IL-17R 抑制剂可能影响 IL-17 通路及神经系统串扰，长期副作用有待商榷，影响市场接受程度和患者顺应性。其余 IL-17 通路的抑制手段以及其临床效果仍有待研究。

从产品竞争角度分析，IL-17A 抑制剂在国内外均有较多的开发者，预计各家药企也会逐步跟进 IL-17A/F 抑制剂的开发。国内进展最快的智翔金泰已经开展了 III 期临床试验。在国外市场而言，目前在研究 IL-17 抑制剂对其它病症的疗效（如克罗恩病等），以及其它类型的 IL-17 抑制剂（如纳米抗体等）。

诺华 IL-17A 单抗已经在市场中取得较为主流的地位，礼来也已经提出了可以小剂量使用的改进 IL-17A 单抗，后续国内产商能否以价格、给药方式、适应症、安全性、有效性等因素取得优势仍然有待观察。

From a development perspective, IL-17A inhibitors were the first among IL-17-targeting agents to reach the market. They have demonstrated good efficacy in treating psoriasis and other autoimmune diseases, with a favourable safety profile and strong druggability.

IL-17A/F dual inhibitors have also achieved excellent results in the treatment of psoriasis, but their application to a broader range of indications remains to be expanded. IL-17R inhibitors may interfere with crosstalk between the IL-17 pathway and the nervous system, raising concerns about potential long-term side effects, which could affect market acceptance and patient compliance. Other strategies for inhibiting the IL-17 pathway and their clinical efficacy still require further investigation.

From a product competition perspective, IL-17A inhibitors have attracted numerous developers both in China and internationally. It is expected that pharmaceutical companies will gradually follow up with the development of IL-17A/F dual inhibitors. In China, Zhixiang Jintai is currently the most advanced, having initiated a Phase III clinical trial. In international markets, research is ongoing into the efficacy of IL-17 inhibitors for other diseases (such as Crohn's disease), as well as the development of alternative formats of IL-17 inhibitors, such as nanobodies.

Novartis' IL-17A monoclonal antibody has already secured a mainstream position in the market, and Eli Lilly has introduced an improved IL-17A antibody designed for use at lower doses. Whether domestic manufacturers can gain a competitive edge through factors such as pricing, mode of administration, indication expansion, safety, and efficacy remains to be seen.

IL-17 背景介绍/ Background

白细胞介素-17 (IL-17) 是促炎性胱氨酸结细胞因子家族。该细胞因子家族主要但不完全是由一组称为 Th17 细胞产生的，以响应其对 IL-23 的刺激。IL-17 与 TNF

Interleukin-17 (IL-17) belongs to the pro-inflammatory cysteine-knot cytokine family. This cytokine family is primarily, though not exclusively, produced by a subset of T cells known as Th17 cells in response to stimulation by IL-23. IL-17 acts synergistically with TNF and IL-1 to promote

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和 IL-1 协同作用促进炎症，诱导了许多免疫信号分子的产生，因此起到重要的免疫调节作用；其中，最显著的作用是其参与诱导和介导 Th17 型促炎反应。

具体来说，在 Th17 型促炎反应中，IL-17 诱导产生来自多种细胞类型（成纤维细胞、内皮细胞、上皮细胞、角质形成细胞和巨噬细胞）的不同细胞因子（例如 IL-6、G-CSF、GM-CSF、IL-1 β 、TGF- β 、TNF- α ）、趋化因子（包括 IL-8、GRO- α 和 MCP-1）和前列腺素（例如 PGE₂）。细胞因子的诱导释放引起气道重塑等常见的 Th17 反应。

除此之外，目前的研究表示 IL-17 功能对于被称为 Th17 细胞的 CD4⁺ T 细胞子集的存续也至关重要。由于这些作用，IL-17 家族与许多与免疫/自身免疫相关的疾病有关，包括类风湿性关节炎、哮喘、狼疮、同种异体移植排斥、抗肿瘤免疫性以及近年来的银屑病和多发性硬化症等[1-2]。

IL17A 蛋白是 IL-17 家族的创始成员，有时也被直接称为 IL17 蛋白，与 T 淋巴细胞型感染性横纹肌病毒 saimiri 基因组编码的病毒类 IL-17 蛋白具有高度同源性，但与其他已知的细胞因子家族没有同源性。

除 IL-17A 以外，IL-17 家族还包括 IL-17B、IL-17C、IL-17D、IL-17E（现在称为 IL-25）和 IL-17F。IL-17 家族的所有成员都具有相似的蛋白质结构，其中研究最为透彻的是 IL-17A 与 IL-17F，二者也为目前临床研究主要靶点。

IL17A 和 IL17F 有相似的生物活性，但是在调节炎症方面以上文依赖的方式发挥不同的活性；二者可以形成同二聚体和 IL-17A–IL-17F 异二聚体。

这些二聚体配体与 IL-17 受体（IL-17R）复合物结合，后者由 IL-17RA 和 IL-17RC 链组成。IL-17A 配体与 IL-17R 结合后激活 IL-17 通路，加剧银屑病等自身免疫疾病，因此靶向阻断 IL-17A 与 IL-17R 的结合或者其上游的通路可以阻断 Th17 型促炎反应的激活，因而有着重要的临床意义[3]。

inflammation by inducing the production of various immune signalling molecules, thereby playing a key role in immune regulation. Its most prominent function lies in initiating and mediating Th17-type pro-inflammatory responses.

Specifically, in Th17 pro-inflammatory responses, IL-17 induces the production of various cytokines, such as IL-6, G-CSF, GM-CSF, IL-1 β , TGF- β , and TNF- α , by a range of cell types including fibroblasts, endothelial cells, epithelial cells, keratinocytes, and macrophages. It also promotes the release of chemokines (including IL-8, GRO- α , and MCP-1) and prostaglandins (such as PGE₂). The release of these cytokines and mediators leads to common Th17-associated responses such as airway remodelling.

In addition, current research indicates that IL-17 plays a crucial role in the survival of Th17 cells, a subset of CD4⁺ T cells. Due to these functions, the IL-17 family has been implicated in a wide range of immune and autoimmune-related diseases, including rheumatoid arthritis, asthma, lupus, allogeneic transplant rejection, anti-tumour immunity, and, more recently, psoriasis and multiple sclerosis.

The IL-17A protein is the founding member of the IL-17 cytokine family and is sometimes simply referred to as IL-17. It shares high homology with a viral IL-17-like protein encoded by the genome of T lymphotropic Herpesvirus saimiri, but it does not exhibit homology with any other known cytokine families.

In addition to IL-17A, the IL-17 family includes IL-17B, IL-17C, IL-17D, IL-17E (now known as IL-25), and IL-17F. All members of the IL-17 family share a similar protein structure. Among them, IL-17A and IL-17F are the most extensively studied and currently serve as the primary targets in clinical research.

IL-17A and IL-17F share similar biological activity, but they exert different effects on inflammation in a context-dependent manner. They can form homodimers as well as IL-17A–IL-17F heterodimers.

These dimeric ligands bind to the IL-17 receptor (IL-17R) complex, which is composed of the IL-17RA and IL-17RC subunits. When the IL-17A ligand binds to the IL-17R complex, it activates the IL-17 signalling pathway, thereby exacerbating autoimmune diseases such as psoriasis. Consequently, targeting and blocking the interaction between IL-17A and IL-17R, or its upstream signalling pathways, can inhibit the activation of Th17-type pro-inflammatory responses, making it of significant clinical relevance [3].

IL-17 信号通路 / Signalling pathways

与其他细胞因子受体不同，IL-17 受体家族的细胞因子信号通过一种独特的适应分子 ACT1，而不是 JAK-STAT 通路。

IL-17 受体细胞通路，首先招募 Act1 启动，Act1 也被认为是 IL-17 受体信号激活必需的分子，Act1 之前被认为是 BAFF 和 CD40 信号的负调节因子，Act1 具有 E3 泛素连接酶活性，在 IL-17 信号传递后，Act1 迅速招募并泛素化 TRAF6，这是另一种 E3 泛素连接酶，与使用 TRAF6 的其他一样，IL-17 通过 I κ B 激酶(IKK)激活和 I κ B α 降解触发 NF- κ B 的激活。

NF- κ B 随后上调 I κ B ζ 和 B 细胞淋巴瘤 3 编码蛋白(Bcl3)的表达，Bcl3 是 NF- κ B 家族的非规范成员，从而促进了各种 IL-17-NF- κ B 驱动的促炎基因的表达。I κ B ζ 在基因和功能上都与银屑病有关连。

在 IL-17A 刺激下，ACT1 启动 TNF 受体相关因子 6(TRAF6)，并通过其 E3 连接酶功能引导 TRAF6 的泛素化转导下游信号。P50/p65 是经典 NF- κ B 信号转导通路的主要组成部分，可导致 G-CSF 以及 CXCL1、CXCL2 和 CXCL8 的表达。

TRAF6 还能促进丝裂原活化蛋白激酶(MAPK)和激活蛋白 1(AP1)通路及 C/EBP β 和 δ 转录因子的激活，反过来，IL-17-NF- κ B 信号转导通路诱导多种抑制 NF- κ B 激活的负反馈回路，包括去泛素化酶或相关因子，包括 A20、Abin-1 和泛素特异性肽酶 25(USP25)。

在角质形成细胞中，最近的研究表明酪氨酸激酶驱动的通路参与了 IL-17 驱动的信号转导。据报道，SYK 激酶与 Act1 和 TRAF6 结合，从而在 NF- κ B 上游发挥作用，导致 CCR6 配体 CCL20 的表达，CCR6 在多数 Th17 型细胞上表达，有可能驱动 Th17 驱动的炎症的前馈回路。

Unlike other cytokine receptors, the IL-17 receptor family transmits signals through a unique adaptor molecule, ACT1, rather than via the conventional JAK-STAT pathway.

In the IL-17 receptor signalling pathway, the adaptor molecule ACT1 is first recruited to initiate the cascade. ACT1 is considered essential for the activation of IL-17 receptor signalling. Previously thought to act as a negative regulator in BAFF and CD40 signalling, ACT1 possesses E3 ubiquitin ligase activity. Upon IL-17 stimulation, ACT1 rapidly recruits and ubiquitinates TRAF6 (a family of E3 ubiquitin ligases), particularly TRAF6. Similar to other pathways involving TRAF6, IL-17 signalling leads to the activation of I κ B kinase (IKK) and the degradation of I κ B α , thereby triggering the activation of NF- κ B.

NF- κ B subsequently upregulates the expression of I κ B ζ and B-cell lymphoma 3-encoded protein (Bcl3), the latter being an atypical member of the NF- κ B family. This promotes the expression of various pro-inflammatory genes driven by the IL-17-NF- κ B signalling axis. I κ B ζ is associated with psoriasis both genetically and functionally.

Upon IL-17A stimulation, ACT1 activates TNF receptor-associated factor 6 (TRAF6) and, through its E3 ubiquitin ligase activity, mediates the ubiquitination of TRAF6 to propagate downstream signalling. P50/p65, the key components of the classical NF- κ B signalling pathway, are then activated, leading to the expression of G-CSF as well as chemokines such as CXCL1, CXCL2, and CXCL8.

TRAF6 also promotes the activation of the mitogen-activated protein kinase (MAPK) and activator protein 1 (AP-1) pathways, as well as the transcription factors C/EBP β and C/EBP δ . In turn, the IL-17-NF- κ B signalling pathway induces several negative feedback mechanisms that inhibit NF- κ B activation. These include the induction of deubiquitinating enzymes and associated regulatory factors, such as A20, Abin-1, and ubiquitin-specific peptidase 25 (USP25).

In keratinocytes, recent studies have shown that tyrosine kinase-driven pathways are involved in IL-17-mediated signal transduction. SYK kinase has been reported to interact with ACT1 and TRAF6, acting upstream of NF- κ B and leading to the expression of CCL20, the ligand for CCR6. Since CCR6 is expressed on most Th17 cells, this may contribute to a feedforward loop that amplifies Th17-driven inflammation.

CSF-1 或 IL-34 的结合诱导 CSF-1R 链二聚化，导致交叉酪氨酸磷酸化和信号分子通过其磷酸酪氨酸结合结构域与受体的直接结合，导致下游相关基因表达，促进细胞的巨噬细胞增殖、分化和存活。

The binding of CSF-1 or IL-34 induces dimerisation of the CSF-1R receptor chains, leading to cross-phosphorylation of tyrosine residues. This enables signalling molecules to bind directly to the receptor via their phosphotyrosine-binding domains, thereby triggering downstream gene expression that promotes macrophage proliferation, differentiation, and survival.

在 T 细胞受体(TCR)和 Dectin 信号通路中，CARD 家族蛋白与 Bcl10 和癌旁淋巴组织淋巴瘤易位蛋白 1(ParacaspaseMalt1)结合，促进 NF- κ B 活化；阻断 Malt1 少部分抑制角质形成细胞的 IL-17 和肿瘤坏死因子 (TNF)，但还需要做更多的研究。

In the T-cell receptor (TCR) and Dectin signalling pathways, CARD family proteins associate with Bcl10 and the paracaspase MALT1 (mucosa-associated lymphoid tissue lymphoma translocation protein 1) to promote the activation of NF- κ B. Partial inhibition of MALT1 has been shown to suppress IL-17 and tumour necrosis factor (TNF) production in keratinocytes, but further research is needed to fully understand this mechanism.

如前所述，IL-17 还通过 Act1 诱导角质形成细胞的增殖信号，这是它在银屑病中的主要影响因素之一，可能也解释了它在皮肤肿瘤发病中的作用。角质形成细胞的增殖是由非典型的 IL-17 介导的 TRAF/ERK 激活途径来促进肿瘤的发生。

As previously mentioned, IL-17 also promotes proliferative signalling in keratinocytes via ACT1, which is one of its key pathogenic mechanisms in psoriasis and may also explain its role in skin tumour development. The proliferation of keratinocytes is driven by an atypical IL-17-mediated TRAF/ERK activation pathway that contributes to tumorigenesis.

IL-17 药物研究 / Drug development

目前市面上有不同药物参与了 IL-17 通路的调节，例如靶向其上游的 IL-12 与 IL-23 等；靶向 IL-17 的药物开发竞争较为激烈。

Currently, various drugs on the market modulate the IL-17 pathway, including those targeting upstream cytokines such as IL-12 and IL-23. Drug development specifically targeting IL-17 itself is highly competitive.

IL-17 抑制剂已经有 3 款（诺华的可善挺、礼来的拓咨、日本麒麟的立美芙）全球上市且均已在中国获批，一款尚在申请批准上市（UCB 的 Bimekizumab，国内尚未获批）。

There are already three IL-17 inhibitors that have been approved globally and in China: Novartis' Cosentyx (secukinumab), Eli Lilly's Taltz (ixekizumab), and Kyowa Kirin's Lumicef (brodalumab). One additional agent, Bimekizumab from UCB, is still undergoing the approval process and has not yet been approved in China.

当前上市的 IL-17 抑制剂以单抗为主，主要针对 IL-17A，但也有部分药企在研究 IL-17A/F 双特异性单抗。

The currently approved IL-17 inhibitors are primarily monoclonal antibodies targeting IL-17A. However, some pharmaceutical companies are also developing dual-specific monoclonal antibodies that target both IL-17A and IL-17F.

从适应症来说，针对银屑病的在研在试药物较多，但也有针对其它自身免疫疾病的临床试验。

In terms of indications, most investigational and experimental drugs target psoriasis, but there are also clinical trials underway for other autoimmune diseases.

据国内银屑病相关流行病学调查，我国银屑病患病率为 0.47%，目前约有 650 万名银屑病患者。其中

According to epidemiological studies on psoriasis in China, the prevalence rate is approximately 0.47%, with an estimated 6.5 million people affected. Among them, 80% to

80%~90%患者是斑块状银屑病，而约 30%的患者是中重度斑块状银屑病，对患者生活质量有很大影响，需要积极治疗。平安证券 2017 年底研报曾指出，以市场占有率最高的药物为参考，我国银屑病单抗治疗市场有望达 300 亿元。

针对银屑病的临床试验结果显示，IL-17A 单抗（诺华的可善挺、礼来的拓咨）取得了相较喜达诺（乌司奴单抗，靶向 IL-12）和修美乐（阿达木单抗，靶向 TNF α ）这类已经上市的针对自免疫其它靶点的生物制剂更好的疗效；但阻断 IL-17 受体的 IL-17R 单抗（日本麒麟的立美芙）在临床试验中引起原因不明的自杀念头，虽然最后仍然获批上市，但在上市后其销售受影响，远不如诺华的可善挺、礼来的拓咨。IL-17A/F 的双特异性抑制剂（UCB 的 Bimekizumab）在针对银屑病的 III 期临床试验[6]中显示出优于诺华的可善挺的疗效，但在国内尚未上市。

此外，国内外均有众多针对 IL-17 的药物在研，但绝大多数都是针对 IL-17A。恒瑞医药、康方生物、君实生物、再鼎医药等众多企业均在此赛道上进行了布局，且大部分已经开始 II 期临床试验。国内进展最快的智翔金泰已经开展了 III 期临床试验。此外，鑫康合与丽珠医药已经开始对国内首款 IL-17A/F 的双特异性抑制剂进行 I 期临床试验。

除了传统的 IL-17A 单抗研发以外，海外药企也在围绕着药物改良与拓展适应症方面积极进行创新。

例如，诺华的可善挺仍然在不断申请与获批各类型适应症，已经批准与准备批准的适应症达到了 23 个，与大部分厂商以以银屑病为核心的 1-2 个疾病的适应症范围形成对比；除此之外，诺华还在针对哮喘、寻常性痤疮、化脓性汗腺炎推出新的新 IL-17A 抑制剂——CJM112，目前尚处于临床 II 期阶段。

除了诺华，德国制药巨头默克也借助合作伙伴 Ablynx 的纳米抗体，共同开发了一款 IL-17A/F 双特异性半衰期延长纳米抗体——索洛奇单抗（M1095，sonelokimab）。目前完成 II 期临床试验，但未公布结果；在一期试验已经证实安全性以及一定疗效[7]。纳

90% have plaque psoriasis, and around 30% suffer from moderate to severe forms, which significantly impact quality of life and require active treatment. A research report by Ping An Securities at the end of 2017 projected that, based on the market share of leading drugs, the market for monoclonal antibody treatments for psoriasis in China could reach 30 billion RMB.

Clinical trial results for psoriasis have shown that IL-17A monoclonal antibodies—such as Novartis' Cosentyx and Eli Lilly's Taltz—achieved better efficacy than other marketed biologics targeting different autoimmune pathways, such as Stelara (ustekinumab, targeting IL-12) and Humira (adalimumab, targeting TNF α). However, the IL-17 receptor-blocking monoclonal antibody Lumicef (brodalumab, developed by Kyowa Kirin) was associated with unexplained suicidal ideation during clinical trials. Although it was eventually approved, its post-marketing sales have been adversely affected and remain significantly lower than those of Cosentyx and Taltz. The dual-specific IL-17A/F inhibitor Bimekizumab (UCB) demonstrated superior efficacy to Cosentyx in Phase III trials for psoriasis [6], but it has not yet been launched in China.

In addition, numerous IL-17-targeting drugs are under development both in China and internationally, with the vast majority focusing on IL-17A. Several Chinese companies—including Hengrui Medicine, Akeso Biopharma, Junshi Biosciences, and Zai Lab—have entered this field, with most candidates already in Phase II clinical trials. Zhixiang Jintai is currently the most advanced domestically, having initiated a Phase III trial. Meanwhile, Xinkanghe and Livzon Pharmaceutical have begun Phase I clinical trials for the first domestically developed IL-17A/F dual-specific inhibitor.

Beyond the traditional development of IL-17A monoclonal antibodies, overseas pharmaceutical companies are actively pursuing innovation through drug optimisation and the expansion of therapeutic indications.

For example, Novartis' Cosentyx continues to seek and obtain approvals for a wide range of indications, with a total of 23 approved or pending indications—far exceeding the 1–2 psoriasis-centred indications typical of most competitors. In addition, Novartis is developing a new IL-17A inhibitor, CJM112, targeting conditions such as asthma, acne vulgaris, and hidradenitis suppurativa. This agent is currently in Phase II clinical trials.

In addition to Novartis, the German pharmaceutical giant Merck has partnered with Ablynx to develop a long-acting, dual-specific nanobody targeting IL-17A/F—Sonelokimab (M1095). It has completed Phase II clinical trials, although the results have not yet been disclosed. Phase I trials have already confirmed its safety and demonstrated a certain level of efficacy [7]. Nanobodies may offer better tissue penetration than traditional antibodies, making

米抗体可能比传统抗体有更好的组织渗透性，因此索洛奇单抗也体现出 IL-17 抑制剂改进疗效方面的一个新的研发方向。

Sonelokimab a promising new direction in the development of more effective IL-17 inhibitors.

投资分析 / Investment analysis

依据医药魔方数据显示，2020 年可善挺的销售额已经达到了 39.95 亿美元，而拓姿的销售额亦已经达到了 7.88 亿美元，且二者的销售额仍在保持快速增长，显示出当前可观的市场规模与容量。

According to data from Pharmcube, Cosentyx generated USD 3.995 billion in sales in 2020, while Taltz reached USD 788 million. Both products continue to show strong growth, reflecting the substantial market size and capacity of the IL-17 inhibitor segment.

但阻断 IL-17 受体的 IL-17R 单抗立美芙在临床试验中引起原因不明的自杀念头，在上市后影响其销售，体现出市场对于安全性的担忧。

However, the IL-17R monoclonal antibody Lumicef, which blocks the IL-17 receptor, was associated with unexplained suicidal ideation during clinical trials. This has affected its sales post-launch, highlighting market concerns regarding safety.

此外，诺华的可善挺已经进入医保；礼来的拓咨虽未进入医保，但主动降价 54%，这也体现了该靶点较为激烈的竞争以及价格战风险。因此，后来的企业应当控制药物成本，在保证疗效与安全性的同时维持降价空间。因此，寻找有低成本优势的标的企业或许是一个有效的投资策略。

In addition, Novartis' Cosentyx has been included in China's national reimbursement drug list, while Eli Lilly's Taltz, though not yet reimbursed, has voluntarily reduced its price by 54%. This reflects the intense competition around this target and the associated risk of price wars. Therefore, later entrants should focus on controlling drug production costs while maintaining efficacy and safety, in order to retain room for price reductions. Identifying companies with a cost advantage may thus represent an effective investment strategy.

另外，对药企而言，IL-17 靶向治疗银屑病已经是一条较为拥挤且已经出现领跑者的赛道，应当避免过度押注。将 IL-17 抑制剂应用在银屑病以外的疾病（例如克罗恩病等其它自身免疫疾病）、改善当前用药方式以提升患者顺应性（例如改用小剂量更低频或者其它更方便的给药方式）或许为有效的差异化竞争策略。后续国内产商能否以这些策略取得优势仍然有待观察。

Moreover, for pharmaceutical companies, IL-17-targeted therapy for psoriasis is already a crowded field with clear market leaders, and over-investment should be approached with caution. Expanding the application of IL-17 inhibitors to other diseases—such as Crohn's disease and other autoimmune conditions—or improving administration methods to enhance patient adherence (e.g. lower doses, less frequent dosing, or more convenient delivery forms) may offer viable strategies for differentiated competition. Whether domestic manufacturers can gain an advantage through such approaches remains to be seen.

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